

# Sonny T. Jones (He/Him/His)

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[LinkedIn](#), [Website](#), [GitHub](#)

## Interests

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Continual Learning, Reinforcement Learning, Artificial Intelligence, Rehabilitation Robotics, Neural Engineering, Video Games, Weightlifting, PC Enthusiast

## Education

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**Doctor of Philosophy, Biomedical Engineering** 2023 - 2028

University of Utah, Salt Lake City, Utah, USA

Advisor: Ashley Dalrymple

**Master of Science, Biomedical Engineering** 2023 - 2026

University of Utah, Salt Lake City, Utah, USA

**Bachelor of Science, Biomedical Engineering** 2021

University of Utah, Salt Lake City, Utah, USA

Thesis: *"Development of Electrocardiographic Measures for Cognitive Load During Prosthesis Use"*

Advisor: Michael Paskett, Gregory Clark

## Experience

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**Graduate Research Assistant** 2023 - Present

Neural Engineering for Rehabilitation Via Electrical Stimulation (NERVES) Lab

Department of Biomedical Engineering, University of Utah, Salt Lake City, Utah, USA

Mentor: Ashley Dalrymple

**Post-Baccalaureate Research Assistant** 2022 - 2023

Utah NeuroRobotics Lab

Department of Electrical and Computer Engineering, University of Utah, Salt Lake City, Utah, USA

Mentor: Jacob George

**Research Analyst/Assistant** 2021 - 2023

Trajectories of Resilience, Community, and Health (TORCH) Lab

Division of Epidemiology, Department of Medicine, University of Utah School of Medicine, Salt Lake City, Utah, USA

Mentor: Mary Jo Pugh

**Undergraduate Research Assistant** 2019 - 2021

Center for Neural Interfaces

Department of Biomedical Engineering, University of Utah, Salt Lake City, Utah, USA

Mentor: Gregory Clark

## Certifications

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|------------------------------------------------------------------------------|------|
| <b>Deep Learning Certificate</b>                                             | 2025 |
| College of Engineering, University of Utah, Salt Lake City, Utah, USA        |      |
| <b>Reinforcement Learning Specialization</b>                                 | 2023 |
| University of Alberta and Alberta Machine Intelligence Institute on Coursera |      |
| <b>Machine Learning Scientist</b>                                            | 2022 |
| DataCamp                                                                     |      |
| <b>Data Science Professional</b>                                             | 2022 |
| DataCamp                                                                     |      |

## Grants

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|---------------------------------------------------------------------------------|------|
| <b>Trainee Professional Development Award</b>                                   | 2025 |
| Society for Neuroscience, San Diego, California, USA                            |      |
| <b>Graduate Student Travel Assistance Award</b>                                 | 2025 |
| The Graduate School, University of Utah, Salt Lake City, Utah, USA              |      |
| <b>Undergraduate Research Opportunity Program Grant</b>                         | 2020 |
| Office of Undergraduate Research, University of Utah, Salt Lake City, Utah, USA |      |
| <b>Office of Undergraduate Research Small Grant</b>                             | 2019 |
| Office of Undergraduate Research, University of Utah, Salt Lake City, Utah, USA |      |
| <b>Undergraduate Research Opportunity Program Grant</b>                         | 2019 |
| Office of Undergraduate Research, University of Utah, Salt Lake City, Utah, USA |      |

## Scholarships

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|---------------------------------------------------------------------------------|-------------|
| <b>Campbell Endowed Fellowship</b>                                              | 2023 - 2026 |
| College of Engineering, University of Utah, Salt Lake City, Utah, USA           |             |
| <b>John C. Jackson Trust Scholarship</b>                                        | 2021        |
| College of Engineering, University of Utah, Salt Lake City, Utah, USA           |             |
| <b>Dee Undergraduate Research Scholarship</b>                                   | 2020        |
| Office of Undergraduate Research, University of Utah, Salt Lake City, Utah, USA |             |
| <b>President's Scholarship</b>                                                  | 2017-2021   |
| University of Utah, Salt Lake City, Utah, USA                                   |             |

## Peer-Reviewed Journal Publications

### Pre-Print

1. MD Paskett, JK Garcia, **ST Jones**, MR Brinton, TS Davis, CC Duncan, JM Cooper, DL Strayer, GA Clark, "Improving Upper-limb Prosthesis Usability: Cognitive Workload Measures Quantify Task Difficulty", *medRxiv*, **2022**.

## Published

1. AN Dalrymple, **ST Jones**, JB Fallon, RK Shepherd, DJ Weber, “Overcoming Failure: Improving Acceptance and Success of Implanted Neural Interfaces”, *Bioelectronic Medicine, Special Issue for Neural Interfaces*, vol 11, **2025**.
2. K North, **ST Jones**, GM Simpson, AN Dalrymple, “Personalized Gait Rehabilitation with Spinal Cord Stimulation and Machine Learning: Recent Advances and Promising Applications”, Invited Review, *Current Opinions in Biomedical Engineering: Bioelectronic Medicine*, vol 34, **2025**.

## In Preparation

1. **ST Jones**, D Precup, PM Pilarski, AN Dalrymple, “Leveraging Predictive Representations of State from General Value Functions to Inform Decision-Making in an Applied Rehabilitation Setting”.
2. CV Ihediwa\*, KJ Valestrino\*, J Hernandez-Bello, Aubrey Andrus, **ST Jones**, AS Gandhi, AN Dalrymple, “Automated identification of spinal vertebrae and spinal cord stimulation electrodes from fluoroscopic images”.
3. GM Simpson, **ST Jones**, WMJ Young, K North, AN Dalrymple, “Biomechanical adjustments of the lower-limbs when approaching new terrains during walking”.
4. WMJ Young, GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple, “Sensor importance for predicting terrains during walking”.

## Conference Proceedings

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### Submitted

1. **ST Jones**, WMJ Young, SD Uhlrich, PM Pilarski, AN Dalrymple, “Continual Reinforcement Learning for Walking Control in a Musculoskeletal Simulation of Hemiparesis”, *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, Edmonton, AB, Canada.
2. GM Simpson, STP Raghu, **ST Jones**, WMJ Young, PM Pilarski, AN Dalrymple, “Learned Predictions from Wearable Sensor Data to Adaptively Switch Between Flat and Staired Terrains”, *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, Edmonton, AB, Canada.
3. WMJ Young, GM Simpson, **ST Jones**, PM Pilarski, AN Dalrymple, “Pseudo-Online Classification of Walking Terrain Using an Ensemble of Support Vector Machines”, *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, Edmonton, AB, Canada.

### Accepted

1. **ST Jones**, AH Le, R Raj, D Sacharny, Z Al-Halah, TC Henderson, S Kundur, “Fine-Tuning Pre-Trained Faster R-CNN to Reduce False Positives in Guardrail Damage Detection from Dashcam Images”, *IEEE Southwest Symposium on Image Analysis and Interpretation*, Santa Fe, New Mexico, USA, Mar **2026**.

## Published

1. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, “Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals”, *Multi-Disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, preprint on arXiv:2507.16983, Dublin, IRL, Jun **2025**.
2. GM Simpson, K North, **ST Jones**, AN Dalrymple, “A Novel Template-Matching Method for Extracting Gait Cycles from Underfoot Pressure Data”, *IEEE International Consortium for Rehabilitation Robotics (ICORR), RehabWeek*, May **2025**.
3. **ST Jones**, GM Simpson, WMJ Young, K North, PM Pilarski, AN Dalrymple, “Comparative Analysis of Temporal-Difference Learning Methods to Learn General Value Functions of Lower-Limb Signals”, *IEEE International Consortium for Rehabilitation Robotics (ICORR), RehabWeek*, May **2025**.

## Other Publications

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### Undergraduate Research Journal

1. **ST Jones**, MD Paskett, GA Clark, CC Duncan, “Development of Electrocardiographic Measures for Cognitive Load During Prosthesis Use”, *Undergraduate Research Journal, University of Utah*, vol 22, **2021**.

## Invited Speeches and Presentations

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1. AN Dalrymple, **ST Jones**, GM Simpson, WMJ Young, “Rewiring Movement, Control of Walking Using Reinforcement Learning”, *Bionic Limbs for Improved Natural Control (BLINC) Lab*, University of Alberta, Edmonton, AB, Canada, Sep. **2025**.

## Oral Presentations

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1. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, “Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals”, *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2025**.
  2. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, “Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals”, *Inaugural Utah AI Summit*, Salt Lake City, UT, USA, Jun **2025**.
    - 3<sup>rd</sup> Place Lightning Talk
  3. **ST Jones**, GM Simpson, WMJ Young, K North, PM Pilarski, AN Dalrymple. “Comparative Analysis of Temporal-Difference Learning Methods to Learn General Value Functions of Lower-Limb Signals”, *IEEE International Consortium for Rehabilitation Robotics (ICORR), RehabWeek*, May **2025**.

4. **ST Jones**, GM Simpson, WMJ Young, K North, PM Pilarski, AN Dalrymple. "Predicting Sensor Signals During Walking Over Different Terrains Using Reinforcement Learning." *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2024**.
5. **ST Jones**, MD Paskett, GA Clark, CC Duncan. "Development of Electrocardiographic Measures for Cognitive Load During Prosthesis Use." *BME Undergraduate Research Symposium*, Salt Lake City, UT, USA, April **2021**.

## Poster Presentations

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1. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, "Predicting Walking Terrains Using Machine Learning and Predictions of Body-Worn Sensor Signals", *Society for Neuroscience*, San Diego, CA, USA, Nov **2025**.
2. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, "Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2025**.
  - Top 12 Poster Grand Rounds
3. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, "Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals", *Inaugural Utah AI Summit*, Salt Lake City, UT, USA, Jun **2025**.
4. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple, "Hierarchical Reinforcement Learning Framework for Adaptive Walking Control Using General Value Functions of Lower-Limb Sensor Signals", *Multi-Disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, Dublin, IRL, Jun. **2025**.
5. **ST Jones**, GM Simpson, PM Pilarski, AN Dalrymple. "Classification of Walking Terrain Using Actual and Predicted Lower-Limb Sensor Signals". *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2025**.
  - Best PhD Poster in Session
6. **ST Jones**, GM Simpson, WMJ Young, K North, PM Pilarski, AN Dalrymple. "Predicting Sensor Signals During Walking Over Different Terrains Using Reinforcement Learning." *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2024**.
  - Top 12 Poster Grand Rounds
7. **ST Jones**, GM Simpson, K North, PM Pilarski, AN Dalrymple. "Predicting Terrain Transitions After Stroke Using Reinforcement Learning Methods". *James R. Swenson, MD Scientific Symposium Day*, Salt Lake City, UT, USA, May **2024**.
8. **ST Jones**, GM Simpson, K North, PM Pilarski, AN Dalrymple. "Predicting Terrain Transitions After Stroke Using Reinforcement Learning Methods". *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2024**.

9. **ST Jones**, MD Paskett, GA Clark, CC Duncan. "Development of Electrocardiographic Measures for Cognitive Load During Prosthesis Use". *BME Undergraduate Research Symposium*, Salt Lake City, UT, USA, April **2021**.

## Conference Abstracts

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### Submitted

1. **ST Jones**, WMJ Young, SD Uhlrich, PM Pilarski, AN Dalrymple, "Recreating Hemiparetic Gait in Musculoskeletal Simulation using Reinforcement Learning", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2026**.
2. GM Simpson, STP Raghu, **ST Jones**, WMJ Young, PM Pilarski, AN Dalrymple, "Adaptively Switching Between Flat and Staired Terrains Using Learned Predictions from Wearable Sensors", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2026**.
3. WMJ Young, GM Simpson, ST Jones, PM Pilarski, AN Dalrymple, "Classification of Walking Terrain using Support Vector Machines: Offline and Pseudo-Online", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2026**.

### Given

4. GM Simpson, K North, **ST Jones**, WMJ Young, PM Pilarski, AN Dalrymple, "Don't Be Trippin': Understanding the Biomechanics of Walking Across Different Terrains", *Society for Neuroscience*, San Diego, CA, USA, Nov **2025**.
5. K North, GM Simpson, **ST Jones**, C Andersen, AN Dalrymple, "Estimating Ground Reaction Force Waveforms from Shank-Mounted Inertial Measurement Unit Using Bidirectional LSTMs", *Society for Neuroscience*, San Diego, CA, USA, Nov **2025**.
6. WMJ Young, GM Simpson, K North, **ST Jones**, PM Pilarski, AN Dalrymple, "Optimizing Sensor Placement for Terrain Prediction During Walking", *Society for Neuroscience*, San Diego, CA, USA, Nov **2025**.
7. CV Ihediwa, KJ Valestrino, **ST Jones**, JH Bello, AN Dalrymple, "Automated Electrode Identification for Spinal Cord Stimulation Using nnU-Net Segmentation", *Society for Neuroscience*, San Diego, CA, USA, Nov **2025**.
8. CV Ihediwa, KJ Valestrino, **ST Jones**, JH Bello, AN Dalrymple, "Image Processing of X-Rays of the Spine and Spinal Cord Stimulation Implants", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2025**.
9. GM Simpson, I Khimach, K North, **ST Jones**, AN Dalrymple, "A Novel Automated Template-Matching Method for Extracting Gait Cycles from Underfoot Pressure Data", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2025**.
10. WMJ Young, GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple, "Optimizing Sensor Placement for Terrain Classification", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2025**.

11. WMJ Young, GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple. "Optimizing Sensor Placement for Terrain Classification", *Inaugural Utah AI Summit*, Salt Lake City, UT, USA, Jun **2025**.
12. CV Ihediwa, KJ Valestrino, **ST Jones**, JH Bello, AN Dalrymple, "Image Processing of X-Rays of the Spine and Spinal Cord Stimulation Implants", *Inaugural Utah AI Summit*, Salt Lake City, UT, USA, Jun **2025**.
13. CV Ihediwa, KJ Valestrino, **ST Jones**, JH Bello, AN Dalrymple, "Image Processing of X-Rays of the Lumbar Spine and Spinal Cord Stimulation Implants", *International Functional Electrical Stimulation Society (IFESS)*, Chicago, IL, USA, May **2025**.
14. GM Simpson, K North, **ST Jones**, AN Dalrymple. "A Novel Template-Matching Method for Extracting Gait Cycles from Underfoot Pressure Data", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2025**.
15. WMJ Young, GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple. "Optimizing Sensor Placement for Terrain Classification", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2025**.
16. CV Ihediwa, KJ Valestrino, **ST Jones**, JH Bello, AS Gandhi, AN Dalrymple, "Image Processing of X-Rays of the Lumbar Spine and Spinal Cord Stimulation Implants", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2025**.
17. CV Ihediwa, KJ Valestrino, **ST Jones**, J Hernandez-Bello, AS Gandhi, AN Dalrymple, "Image Processing of X-Rays of the Lumbar Spine & Spinal Cord Stimulation Implants", *Undergraduate Research Symposium*, Salt Lake City, UT, USA, April **2025**.
18. CV Ihediwa, KJ Valestrino, **ST Jones**, CT Stanley, AN Dalrymple, "Image Processing of X-Rays of the Lumbar Spine & Spinal Cord Stimulation Implants", *Biomedical Engineering Society (BMES)*, Baltimore, MD, USA, Oct **2024**.
19. GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple, "Finding Optimal Sensor Combinations Across Variable Terrains Using tSNE And Reinforcement Learning", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2024**.
20. CV Ihediwa, KJ Valestrino, **ST Jones**, CT Stanley, AN Dalrymple, "Image Processing of X-rays of the Lumbar Spine and Spinal Cord Stimulation Implants", *Utah Biomedical Engineering Conference (UBEC)*, Salt Lake City, UT, USA, Sep **2024**.
21. GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple, "Optimal Body-Worn Sensors for Predicting Terrain Transitions While Walking", *James R. Swenson Scientific Symposium Day*, Salt Lake City, UT, USA, May **2024**.
22. GM Simpson, **ST Jones**, K North, PM Pilarski, AN Dalrymple, "Optimal Body-Worn Sensors for Predicting Terrain Transitions While Walking", *Rocky Mountain American Society of Biomechanics (RMASB)*, Estes Park, CO, USA, April **2024**.

## Teaching and Lectures

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### Teaching

- **Graduate Teaching Assistant, BME 3101 Biosignals Analysis** 2025  
University of Utah, Salt Lake City, Utah, USA

### Guest Lecturer

- **Graduate Student Panel** 2024  
BME 1010 Careers in Biomedical Engineering, University of Utah, Salt Lake City, Utah, USA
- **“Rehabilitative Robotics and Its Application in Stroke/Amputee Populations”** 2024  
Robotics, West High School, Salt Lake City, Utah, USA
- **“Rehabilitative Robotics and Its Application in Stroke/Amputee Populations”** 2024  
Principles In Engineering, West High School, Salt Lake City, Utah, USA

## Student Mentorship

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- **Addarsh Kuberan** 2025 - Present  
Undergraduate Student, Department of Electrical and Computer Engineering and Department of Physics and Astronomy, University of Utah, Salt Lake City, UT, USA  
Role: Graduate Student Mentor
- **Chimdi Ihediwa** 2024 - 2025  
Undergraduate Student, Department of Biomedical Engineering, University of Utah, Salt Lake City, UT, USA  
Role: Graduate Student Mentor
- **Wyatt Young** 2024 - 2025  
Undergraduate Student, Department of Biomedical Engineering, University of Utah, Salt Lake City, UT, USA  
Role: Graduate Student Mentor

## Open-Source Software and Data Repositories

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1. [BlyncsySFT](#)  
Fine-tuning package for Faster Region-based Convolutional Neural Network (R-CNN) models.
2. [Machine Learning Gait Front End](#)  
Custom graphical user interface (GUI) and control software for synchronized data collection from Delsys Trigno EMG and XSensor wireless pressure insoles.

## Academic Honors/Awards

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- Phi Eta Sigma Honors Society, University of Utah 2017  
Dean's List, University of Utah 2017 - 2021



## Service and Volunteerism

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### University

- **Chair, Curriculum Committee** 2025 - Present  
Biomedical Engineering Graduate Student Advisory Committee (GSAC), University of Utah, Salt Lake City, Utah, USA
- **Volunteer, Inaugural Utah AI Summit** 2025  
John and Maria Price College of Engineering, University of Utah, Salt Lake City, Utah, USA
- **Volunteer, Biomedical Engineering Recruitment Weekend** 2025  
Department of Biomedical Engineering, University of Utah, Salt Lake City, Utah, USA
- **Volunteer, National Biomechanics Day** 2024  
University of Utah, Salt Lake City, Utah, USA
- **Volunteer, Biomedical Engineering Recruitment Weekend** 2024  
Department of Biomedical Engineering, University of Utah, Salt Lake City, Utah, USA
- **External Vice President** 2019 - 2020  
Vietnamese American Student Association (VASA), University of Utah, Salt Lake City, Utah, USA

### Community

- **Judge Advisor Assistant** 2025  
Utah Regional First Robotics Competition, Salt Lake City, Utah, USA
- **Software Mentor** 2023 - 2024  
First Robotics Competition, Red Rock Robotics, West High School, Salt Lake City, Utah, USA
- **Director of Patient Impact** 2022 - 2023  
Project Embrace, West Jordan, Utah, USA
- **Member, Research Committee** 2021 - 2023  
Project Embrace, West Jordan, Utah, USA
- **Director of Marketing and Community Engagement** 2021 - 2022  
Project Embrace, West Jordan, Utah, USA
- **Community Outreach and Engagement Chair** 2020 - 2021  
Southwest Union of Vietnamese Student Associations (SWUVSA)
- **Counsel of School Representatives (CoSR) Member** 2019 - 2020  
Southwest Union of Vietnamese Student Associations (SWUVSA)
- **Emergency Room Volunteer** 2019 - 2020  
University of Utah Hospital, Salt Lake City, Utah, USA
- **Medical Interpreter & Patient Assistance Program** 2019 - 2020  
Maliheh Free Clinic, Salt Lake City, Utah, USA
- **Camp Counselor** 2018, 2019  
Camp Anytown, Inclusion Center, Salt Lake City, Utah, USA

## Media

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### Television

- [“Medical Innovations at the University of Utah”](#) 2022  
University of Utah, The College Tour, Amazon

### Print

- [“AI Summit: Cutting-Edge Talks From Across Campus and Beyond”](#) 2025  
John and Maria Price College of Engineering, University of Utah, Salt Lake City, Utah, USA